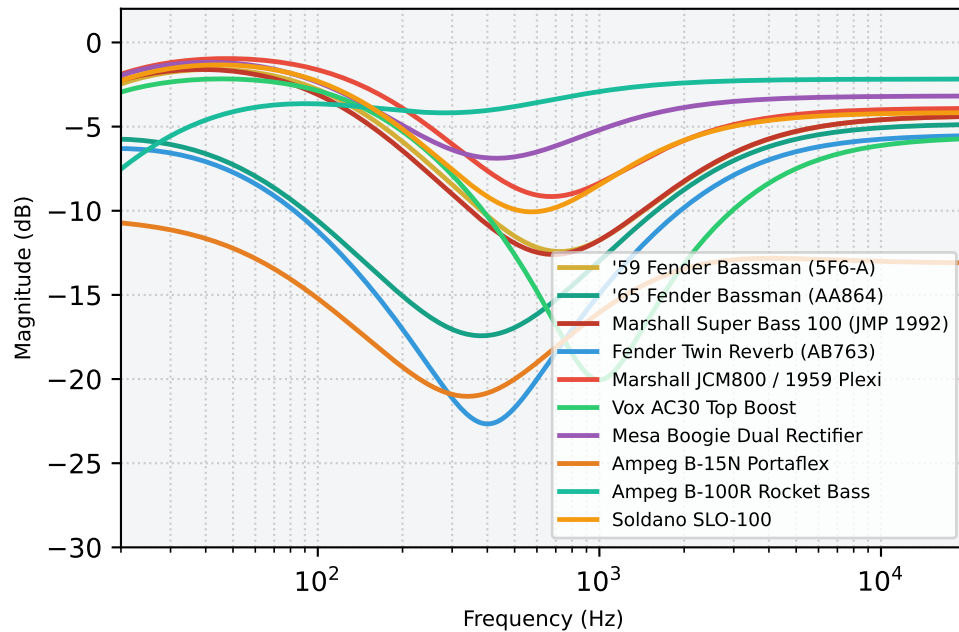


# CLASSIC AMP TONE STACK COMPARATIVE BENCHMARK

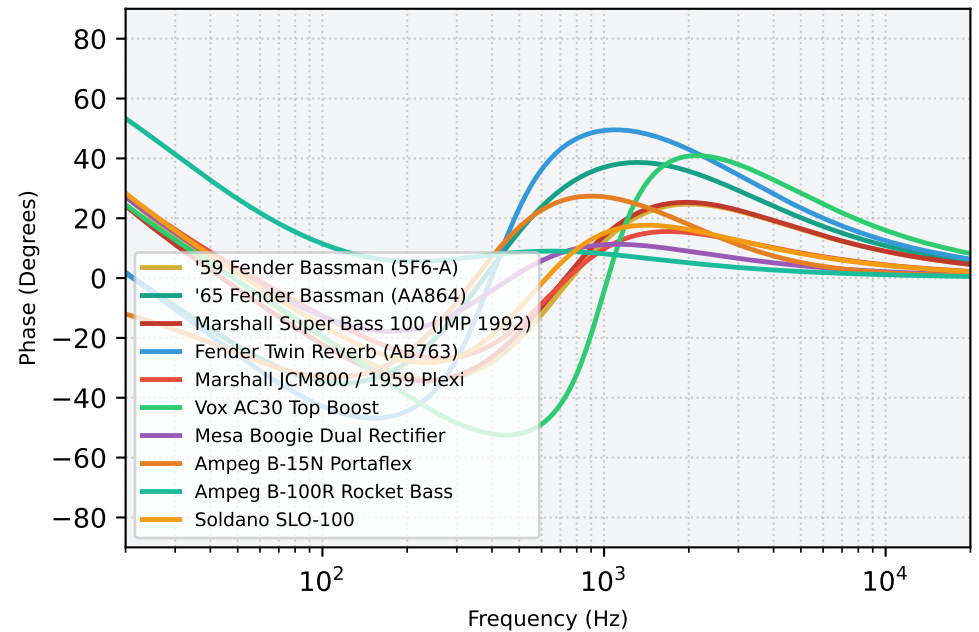
Mathematical Comparison of 10 Iconic Circuit Topologies (Fender, Marshall, Ampeg, Vox, Mesa, Soldano)

Yeh & Smith DAFx-06 Nodal Synthesis

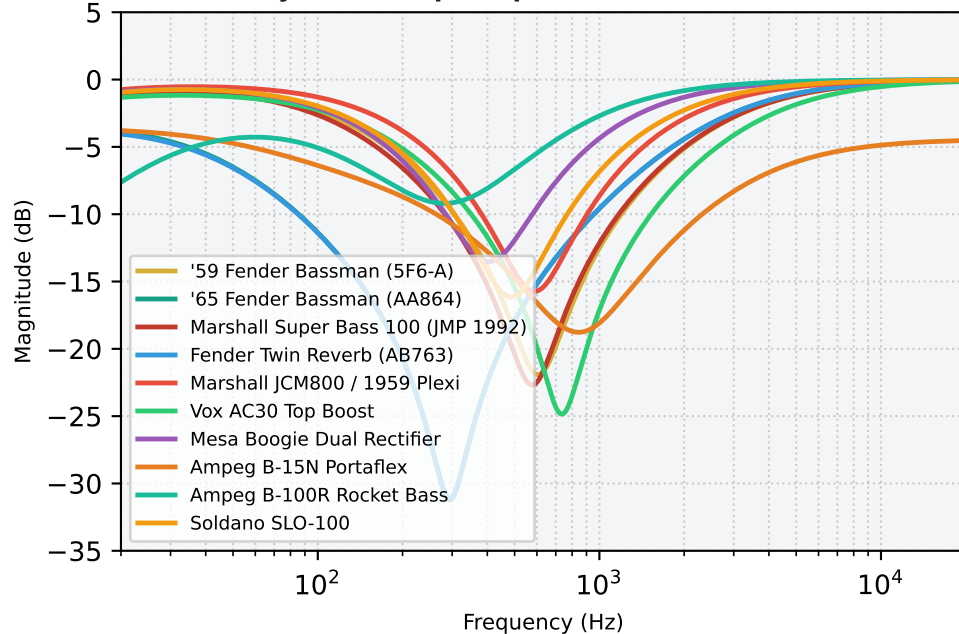
Factory Default Tone Signatures (T=5, M=5, B=5)



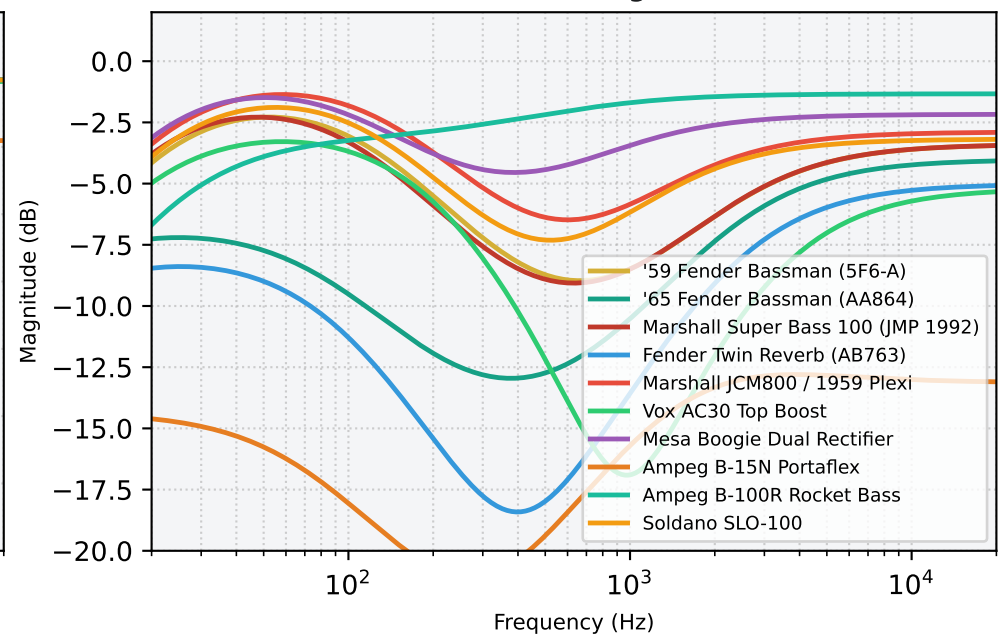
Phase Trajectories across Tone Stacks



Heavy Mid-Scoop Response (T=10, M=0, B=10)



Mid Push / Solo Lead Setting (T=5, M=10, B=3)



# CIRCUIT COMPONENT SPECIFICATIONS & ACOUSTIC MAPPING

Mathematical Component Values and Characteristic Notch Frequencies

Ready for C++ Modular Implementation

|          | Amplifier Circuit            | R1 (Treb) | R2 (Bass) | R3 (Mid) | R4 (Slope) | C1 (Treb) | C2 (Bass) | C3 (Mid) | Notch Freq | Acoustic Personality                                                      |
|----------|------------------------------|-----------|-----------|----------|------------|-----------|-----------|----------|------------|---------------------------------------------------------------------------|
| Marshall | '59 Fender Bassman (5E6)     | 250k      | 1.0M      | 25k      | 56k        | 250 pF    | 20 nF     | 20 nF    | 725 Hz     | Original Tweed benchmark; balanced warmth, 80dB gain                      |
|          | '65 Fender Bassman (A58)     | 250k      | 250k      | 25k      | 100k       | 250 pF    | 100 nF    | 47 nF    | 380 Hz     | Blackface Bassman; 100nF deep bass cap, 47nF mid cap                      |
|          | Marshall Super Bass 100 (J1) | 250k      | 1.0M      | 25k      | 56k        | 250 pF    | 22 nF     | 22 nF    | 585 Hz     | Heavy tube bass; 56k slope, 250pF treble, 22nF bass                       |
|          | Fender Twin Reverb (A58)     | 250k      | 250k      | 10k      | 100k       | 250 pF    | 100 nF    | 47 nF    | 400 Hz     | Classic Blackface; huge low-end authority, deeper scooped                 |
|          | Marshall JCM800 / 1959       | 220k      | 1.0M      | 22k      | 33k        | 470 pF    | 22 nF     | 22 nF    | 672 Hz     | British crunch; forward midrange, higher treble bite                      |
|          | Vox AC30 Top Boost           | 1000k     | 1.0M      | 10k      | 100k       | 47 pF     | 22 nF     | 10 nF    | 908 Hz     | British chirp; small treble cap, 100k slope, spiky                        |
|          | Mesa Boogie Dual Rect        | 250k      | 1.0M      | 50k      | 47k        | 680 pF    | 20 nF     | 20 nF    | 420 Hz     | Modern metal; 680pF treble cap, 50k mid pot for extreme                   |
|          | Ampeg B-15N Portaflex        | 470k      | 1.0M      | 10k      | 100k       | 330 pF    | 0 nF      | 33 nF    | 341 Hz     | Legendary tube bass; Baxandall stack; flat center, independent            |
|          | Ampeg B-100R Rocket          | 50k       | 50k       | 50k      | 33k        | 4700 pF   | 33 nF     | 47 nF    | 284 Hz     | Iconic solid-state/tube emulator; 284Hz combo with Ultra-Mid contour; 50k |
|          | Soldano SLO-100              | 250k      | 1.0M      | 25k      | 47k        | 470 pF    | 20 nF     | 20 nF    | 579 Hz     | Boutique high-gain; legendary singing lead, smooth                        |